

Note on: *Treasury Lock Model*

Anonymous model documentation

Reading notes

May 24, 2026

Metadata

Title	Treasury Lock Model
Author(s)	Not stated
Date	Not stated
Source	Document references finpricing.com ; no SSRN/DOI
Local file	<code>papers/anon_treasury_lock_model/31670362145195.pdf</code>
Tags	<code>treasury, t-lock, forward, repo, pv01, model-doc</code>
Date read	May 24, 2026

One-paragraph summary

A four-page model-documentation note describing the implementation of a T-Lock pricer. It states the bond forward price as a clean-price formula discounted with the repo curve, then writes the T-Lock NPV in three equivalent forms depending on what is locked: the clean price, the yield (via PV01), or the dirty price. The yield-lock form uses a discrete two-sided PV01 around the forward yield rather than a closed-form derivative. The document closes with a caveat that PV01 and NDELTA are underlier-related and only approximate the T-Lock’s own sensitivities.

1 Setup and notation

The document mixes price-quote conventions (clean/dirty/forward) with yield-locked conventions. It uses one repo discount factor for the forward-price construction and a separate “bond zero curve” discount factor for the T-Lock NPV. This dual-curve structure is not typical of more academic treatments (e.g. Pucci 2019, where a single risk-free DF D_{t,t_e} is used and the repo enters only in the forward drift).

Document	House	Meaning
P_0	$P_t^{mkt, clean}$	Clean price of the underlying bond at spot
A_0, A_{del}	—	Accrued interest at spot, at delivery
T_0, T_{del}	t, t_e	Spot settlement, forward settlement (delivery) dates
N	—	Number of coupons between T_0 and T_{del}
C_i, T_i	c_i, t_i	i -th coupon and its payment date
B_f	$\text{Fwd}_t(t_e)$ (clean)	Bond forward (clean) price at delivery
$D_{repo}(\cdot, \cdot)$	—	Discount factor from the repo zero curve
$D_{\text{T-Lock}}(\cdot, \cdot)$	—	Discount factor from the bond zero curve (used to discount T-Lock NPV)
M	N (notional)	Trade amount
y_f	IRR_{fwd}	Forward bond yield at delivery
$y_{\text{T-Lock}}$	L	Locked yield
$P_{\text{T-Lock}}$	$L^{(P)}$	Locked clean price
$PV01(y_f)$	—	Two-sided finite-difference PV01 at y_f
$B(y)$	$P_t(y)$	Bond PV at yield y

2 Key equations

Bond forward (clean) price.

$$B_f = \frac{P_0 + A_0}{D_{repo}(T_0, T_{del})} - \sum_{i=1}^N \frac{C_i}{D_{repo}(T_i, T_{del})} - A_{del}. \quad (1)$$

Says: accrete the dirty spot price ($P_0 + A_0$) to delivery at the repo rate, subtract coupon flows that occur in $[T_0, T_{del}]$ each accreted to delivery, then strip the delivery-date accrued interest to land on a *clean* forward. This is the same content as Pucci’s eq. (49) in continuous form, here in discrete-DF form, and expressed clean instead of dirty.

T-Lock NPV: clean-price lock.

$$\text{NPV}_{\text{T-Lock}} = N \cdot (P_{\text{T-Lock}} - B_f) \cdot D_{\text{T-Lock}}(t, T_{del}). \quad (2)$$

Says: a forward contract on the clean price. The reference forward B_f is computed from (1) using the repo curve; the resulting payoff is then discounted to t using a separate “bond zero” discount curve.

T-Lock NPV: yield lock (via PV01).

$$\text{NPV}_{\text{T-Lock}} = M \cdot (y_f - y_{\text{T-Lock}}) \cdot PV01(y_f) \cdot D_{\text{T-Lock}}(t, T_{del}), \quad (3)$$

with the PV01 defined as a two-sided finite difference

$$PV01(y_f) = B(y_f + 0.5 \text{ bp}) - B(y_f - 0.5 \text{ bp}). \quad (4)$$

Says: the linear payoff $(y_f - L) \cdot PV01$ is the discrete analogue of Pucci’s $\text{RiskFactor}_{t_e} \cdot (\text{IRR}_{t_e} - L)$. The sign convention here makes $PV01(y_f)$ *positive* (because a positive yield shift lowers price, but the formula computes $B(y_f +) - B(y_f -)$ with B decreasing in y , giving a negative quantity — see commentary).

T-Lock NPV: dirty-price lock. The document states that the dirty case is obtained from the clean-price case via $P^{\text{dirty}} = P^{\text{clean}} + A$, i.e. it computes the clean-price lock and adjusts by accrued interest. No new formula.

3 Derivations & commentary

Commentary

The PV01 sign convention in (4) as written gives a *negative* number (because B is decreasing in y). If one takes the document's formula literally then a long T-Lock with $y_f > y_{\text{T-Lock}} = L$ produces a negative NPV, which contradicts the intent (long T-Lock pays when yields rise). The conventional practice is either (a) define PV01 as the absolute price change per +1 bp, or (b) flip the sign of $(y_f - L)$. The document does not say which it adopts. House interpretation: take $\widetilde{PV01}(y_f) = -[B(y_f + 0.5\text{bp}) - B(y_f - 0.5\text{bp})] > 0$, which matches Pucci's $\text{RiskFactor}_{t_e} = -\partial P(y)/\partial y > 0$ in the limit of small bump width.

Commentary

The document uses two distinct discount curves: a repo curve for the forward bond price, and a “bond zero curve” for the T-Lock NPV. The choice of the latter is unusual; Pucci (2019) and standard post-2008 practice would use a collateral/OIS rate for derivative discounting, not a bond curve. This is the kind of detail that should be confirmed against the firm's actual CSA before using the formula.

Commentary

The finite-difference PV01 in (4) differs from a closed-form $-\partial P(y)/\partial y$ by terms of order bp^2 . For practical purposes (1 bp is small) the difference is negligible, but a property test should compare the two and check that the finite-difference converges to the analytic derivative as the bump shrinks.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- B_f in (1), the clean forward bond price constructed from the repo-curve discount factors.
- $\text{NPV}_{\text{T-Lock}}$ in three equivalent formulations: clean-price lock (2), yield lock (3), dirty-price lock (derived).
- $PV01(y_f)$ as a two-sided finite difference (4).

Canonical numerical case: The document provides no worked example. For validation purposes, re-use the Pucci 2019 canonical case (US Treasury 3.125% 25-Nov-2028, ISIN US9128285M81, 24-Jan-2019, $t_e - t = 3$ months, $P^{\text{mkt}, \text{dirty}} = 104.1055$, $\text{IRR} = 2.717\%$, $r^{\text{repo}} = 2.46\%$); the two documents should agree to first order.

Expected numerical values: not available in this document. Cross-check is against the Pucci numbers (B_f dirty = 104.74; forward $\text{IRR} = 2.53\%$) within ± 0.01 and ± 1 bp.

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(1)	B_f	clean forward bond price via repo discount factors
(2)	$NPV_{T\text{-Lock}}$	T-Lock NPV when the clean price is locked
(3)	$NPV_{T\text{-Lock}}$	T-Lock NPV when the yield is locked, via PV01
(4)	$PV01(y_f)$	two-sided finite-difference PV01 at the forward yield

Invariants and identities to verify:

1. *Equivalence of clean and dirty locks*: the dirty-price formulation, after subtracting A_{del} from the locked price, equals the clean-price formulation (2).
2. *First-order agreement with the analytic derivative*: as the FD bump width $\rightarrow 0$, $PV01(y_f)/bp$ converges to $-\partial P(y)/\partial y|_{y=y_f}$.
3. *First-order agreement of yield and clean-price locks*: for $y_{T\text{-Lock}}$ near y_f , the clean-price and yield formulations agree to $O((y_f - y_{T\text{-Lock}})^2)$ when the locked clean price is $P_{te}(y_{T\text{-Lock}})$.
4. *Repo-curve consistency*: B_f as constructed in (1) satisfies the no-arbitrage condition that a repo strategy returns zero P&L when valued at T_{del} .

5 Glossary of symbols (paper's notation)

P_0	Clean price of the underlying bond at spot
A_0	Accrued interest on a unit underlying bond at T_0
T_0	Spot settlement date
T_{del}	Delivery date (forward settlement date)
A_{del}	Accrued interest on a unit underlying bond at T_{del}
N (eq. 1)	Number of coupons paid in $[T_0, T_{del}]$
N (eq. 2)	Quantity of the underlying bond (notional)
C_i	i -th coupon of the unit underlying bond
T_i	i -th coupon payment date
$D_{repo}(t_1, t_2)$	Discount factor from t_2 back to t_1 , repo zero curve
$D_{T\text{-Lock}}(t_1, t_2)$	Discount factor from t_2 back to t_1 , bond zero yield curve
B_f	Bond forward clean price at delivery
$P_{T\text{-Lock}}$	Locked clean price for the underlying bond
M	Trade amount of the underlying bond
y_f	Forward bond yield (as if bond settles at delivery)
$y_{T\text{-Lock}}$	Locked bond yield
$PV01(y_f)$	$B(y_f + 0.5 \text{ bp}) - B(y_f - 0.5 \text{ bp})$
$B(y)$	Present value of the underlying bond given yield y
t	Valuation date

TODO / Open questions

- Confirm the PV01 sign convention used in eq. 3 — as written, the signs of $(y_f - L)$ and the FD-PV01 do not produce the conventional “long T-Lock pays on yield rise” behaviour.
- The document uses a separate “bond zero curve” for discounting the T-Lock NPV.

Confirm against firm CSA whether this is appropriate; modern practice would normally use OIS/collateral.

- No worked numerical example in the document. Cross-validate (1) against the Pucci 24-Jan-2019 case.